

Basic Details Of The Problem Statement

Hackathon name : BFSI Launchpad 2025

Problem Statement Title: AI-Powered Insurance Policy Information Chatbot

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Theme Name: Smart Automation using AI in BFSI sector



Problem Statement Details

Background: Insurance companies offer a variety of policies, including health, life, auto, and home insurance. Customers often have questions about the different types of policies, coverage options, premiums, and claim processes. Providing timely and accurate information is crucial for customer satisfaction and decision-making. An AI-powered chatbot can assist customers by answering their queries about insurance policies, helping them understand their options and make informed decisions.

Objective: Develop a chatbot that can assist customers with queries related to different types of insurance policies offered by the company. The chatbot should understand natural language, provide accurate responses, and guide customers through the information they need.

Requirements:

LLM: Use LLM for natural language processing

Knowledge Base Integration: Create a knowledge base with information about different insurance policies, coverage options, premiums, and claim processes. Details can be shared in the form of PDF to create the knowledge base.

Data contextualization: Ensure the chatbot can retrieve and present relevant information from the knowledge base.

Conversation Management: Implement fallback mechanisms to escalate complex issues to human agents if necessary. they need.

Current Status:

Working model is ready for demonstration , further build is on progress. **Testing** and **validation** process are next to be undergone.

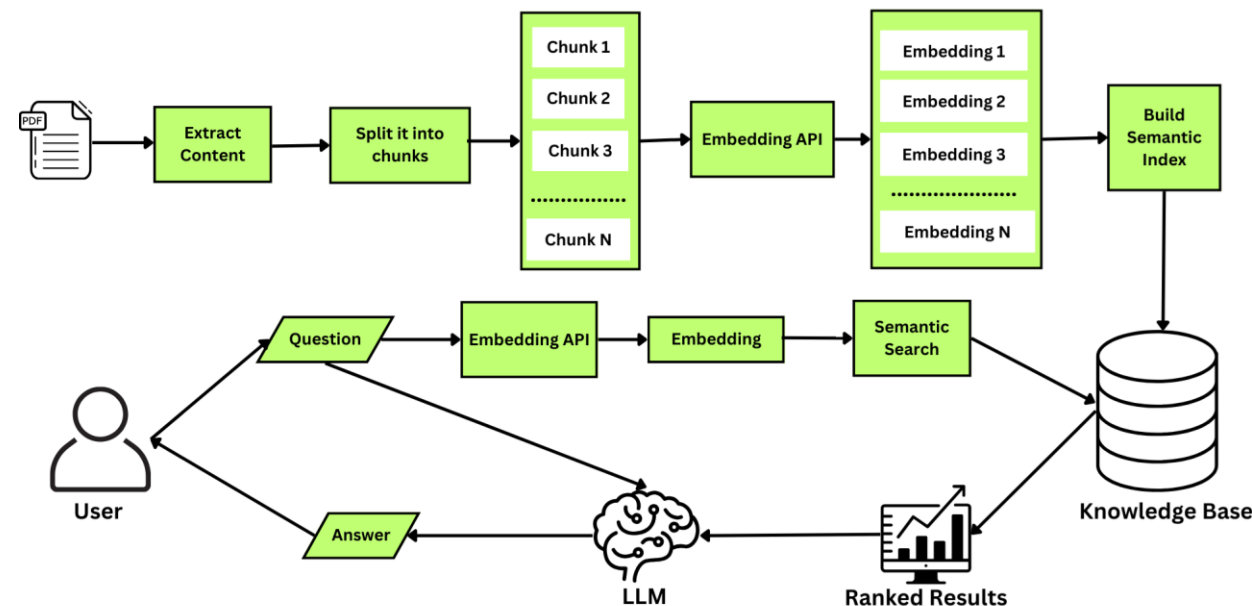


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Idea/Approach Details

Idea/Solution/Prototype

- **Leverage opensource technology**: To reduce **development costs**, and benefit from a wide range of freely available tools, **libraries**, and resources in building the chatbot.
- **Contextualized Conversation**: We are creating **NLP-driven** chatbot with **context management**, intent recognition, and user engagement ensuring to the point answers.
- **Query Categorization**: Categorize queries based on the type of information users seek (e.g., regulatory queries, land-related queries, safety compliance).
- **Retrieval-Augmented Generation (RAG)**: Incorporated a hybrid approach by combining semantic search with large language models. This ensures precise and grounded responses based on actual documents, reducing hallucinations from the LLM.5.
- **Fallback Mechanism with LLM**: Integrated a smart fallback system that triggers a direct LLM response when the retrieval-based answer is vague or insufficient, ensuring that users always receive a meaningful and informative reply.



Technology stack :

Component	Tool / Library
Language	Python
PDF Processing	LangChain + PyPDFLoader
Chunking	RecursiveCharacterTextSplitter
Embeddings	GoogleGenerativeAIEmbeddings
Vector DB	FAISS
LLM	Gemini 2.0 Flash (Google)
UI	Command-line + IPython.display

Idea/Approach Details

Use Cases :

- **General Purpose Usage:** The chatbot can give answers about general knowledge questions accurately and with updated information.
- **Overcoming Language Barriers:** A multilingual web based platform, based on **32 different languages**, which will help our users to communicate with our **chatbot**.
- **Information Retrieval:** The chatbot can provide **up-to-date concise summaries of the relevant sections**.
- **Natural Language Processing (NLP) Model Training:** Training the NLP model to understand user intents, classify queries, and generate appropriate responses.

Outcomes

- ✓ Accurate policy information retrieval
- ✓ Robust fallback logic
- ✓ Efficient document handling with FAISS
- ✓ Ready for extension to Streamlit Web UI

Dependencies / Show stopper :

- **Data Availability and Quality:** Availability of comprehensive and **up-to-date information** is difficult.
- **Hardware Limitations:** Using advanced models(**ChatGPT, LLama**), to train or deploy requires heavy Computing power.
- **API Limitations:** for deploying LLM models locally or in cloud we have to use **API keys**. Which can be prone to **server errors and network issues**.
- **Token Limit:** There is **token limitation** in every model which affects the length and quality of the output produced.

Channels & Revenue Streams:

- Govt portals , Pvt industries, organizations etc.
- Subscription Model, Pay-Per-Use, Enterprise Licensing.



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